

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0136

Basilone 138 kV
June 01, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0136. The Interconnection Customer proposes a 562.0 MW natural gas combined cycle facility interconnecting at the Basilone 138 kV substation in the West Virginia area. The request was received on June 10, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 90 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$161,001,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Angel Mounds – Butterfield 138 kV Line	Faraday – Basilone 138 kV	277.5	200	100.0
Faraday – Far Rockaway 138 kV L	Butterfield – Basilone Substation 138 kV	268.7	207	100.0
Faraday – Basilone Substation 138 kV L	Basilone – Butterfield 138 kV	265.9	210	100.0
Butterfield – Basilone 138 kV	Basilone Substation – Faraday 138 kV	261.7	214	100.0
King – Angel Mounds 138 kV L	Faraday – Basilone 138 kV	249.9	215	100.0
Lyon County – Far Rockaway 138 kV	Butterfield – Basilone 138 kV	161.4	227	72.0
Dooley – Lyon County 138 kV	Basilone Substation – Butterfield 138 kV	157.8	230	72.0
King – Claypool Hill 138 kV	Faraday – Basilone 138 kV	145.4	204	63.6
Coal Creek – Claypool Hill 138 kV	Faraday – Basilone 138 kV	130.7	218	63.6
Baileysville – Mullensville 138 kV	Far Rockaway Substation – Lyon County 138 kV	120.4	209	47.8
Aspen 345/138 kV Auto	Crandall – Aspen 345 kV line	120.4	407	8.6
Mullensville – Shiloh 138 kV	Lyon County – Far Rockaway 138 kV	118.4	217	47.8
Shiloh – Welch Substation 138 kV	Lyon County – Far Rockaway 138 kV Line	116.7	224	47.8
Far Rockaway – Dry Fork 138 kV Line	Far Rockaway – Lyon County Substation 138 kV	115.3	235	47.8
Welch – Dry Fork 138 kV line	Far Rockaway – Lyon County 138 kV	115.2	231	47.8
Dayton Substation – Harrisonburg 138 kV L	Chatom – Dayton 345 kV	112.4	377	5.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hilltop 500/345 kV Auto	Hilltop – Talenenergy Martins Creek Llc Williamsport 500 kV	110.7	806	5.1
Ft Humbug Substation – West Campus 345 kV	Orchard Road – Fort Hubug 500 kV	109.7	1,371	6.0
Kean Mtn And Grassy Creek 138 kV/1 kV Transformer Bank	Kean Mtn And Grassy Crk – Clinch Riv 345 kV Line	101.3	296	24.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Angel Mounds – Butterfield 138 kV	Rebuild line with higher-capacity conductor	\$18,849,000	100% (\$18,849,000)
Faraday – Far Rockaway 138 kV L	Rebuild line with higher-capacity conductor	\$10,423,000	100% (\$10,423,000)
Faraday – Basilone Substation 138 kV L	Rebuild line with higher-capacity conductor	\$5,156,000	100% (\$5,156,000)
Butterfield – Basilone 138 kV	Rebuild line with higher-capacity conductor	\$13,199,000	100% (\$13,199,000)
King – Angel Mounds 138 kV L	Rebuild line with higher-capacity conductor	\$15,647,000	100% (\$15,647,000)
Lyon County – Far Rockaway 138 kV	Rebuild line with higher-capacity conductor	\$17,916,000	100% (\$17,916,000)
Dooley – Lyon County 138 kV	Rebuild line with higher-capacity conductor	\$7,148,000	100% (\$7,148,000)
Claypool Hill – King 138 kV	Rebuild line with higher-capacity conductor	\$2,193,000	99% (\$2,165,000)
Coal Creek – Claypool Hill 138 kV	Rebuild line with higher-capacity conductor	\$10,896,000	99% (\$10,755,000)
Baileysville – Mullensville 138 kV	Rebuild line with higher-capacity conductor	\$8,632,000	100% (\$8,632,000)
Aspen 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$17,892,000	29% (\$5,125,000)
Mullensville – Shiloh 138 kV	Rebuild line with higher-capacity conductor	\$5,558,000	100% (\$5,558,000)
Shiloh – Welch Substation 138 kV	Rebuild line with higher-capacity conductor	\$5,185,000	100% (\$5,185,000)
Dry Fork – Far Rockaway 138 kV	Rebuild line with higher-capacity conductor	\$4,595,000	100% (\$4,595,000)
Welch – Dry Fork 138 kV line	Rebuild line with higher-capacity conductor	\$12,578,000	100% (\$12,578,000)
Dayton – Harrisonburg 138 kV	Rebuild line with higher-capacity conductor	\$1,633,000	100% (\$1,633,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hilltop 500/345 kV Transformer	Replace transformer with higher-rated unit	\$27,381,000	36% (\$9,757,000)
Ft Humbug Substation – West Campus 345 kV	Reconductor line	\$11,899,000	27% (\$3,256,000)
Kean Mtn And Grassy Crk 138 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$5,569,000	62% (\$3,424,000)
	Total allocated to project		\$161,001,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.